

Futureproof

EPM v14a

Scenario Analysis

Loan Term Mix & Top Cover Limit — 50,000-Path Monte Carlo
March 2025

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Executive Summary

This report analyses two portfolio-level parameter changes to the FutureProof EPM model, using independent 50,000-path Monte Carlo simulations for each loan tenure (15, 20, 25, and 30 years). All simulations use v14a parameters with identical market assumptions.

Scenario 1 — Loan Term Mix: Removing 15-year terms and redistributing equally across 20, 25, and 30-year tenures reduces weighted portfolio PoD by **2.4 percentage points** (from 20.0% to 17.6%) and increases mean surplus by **\$189K per mortgage**.

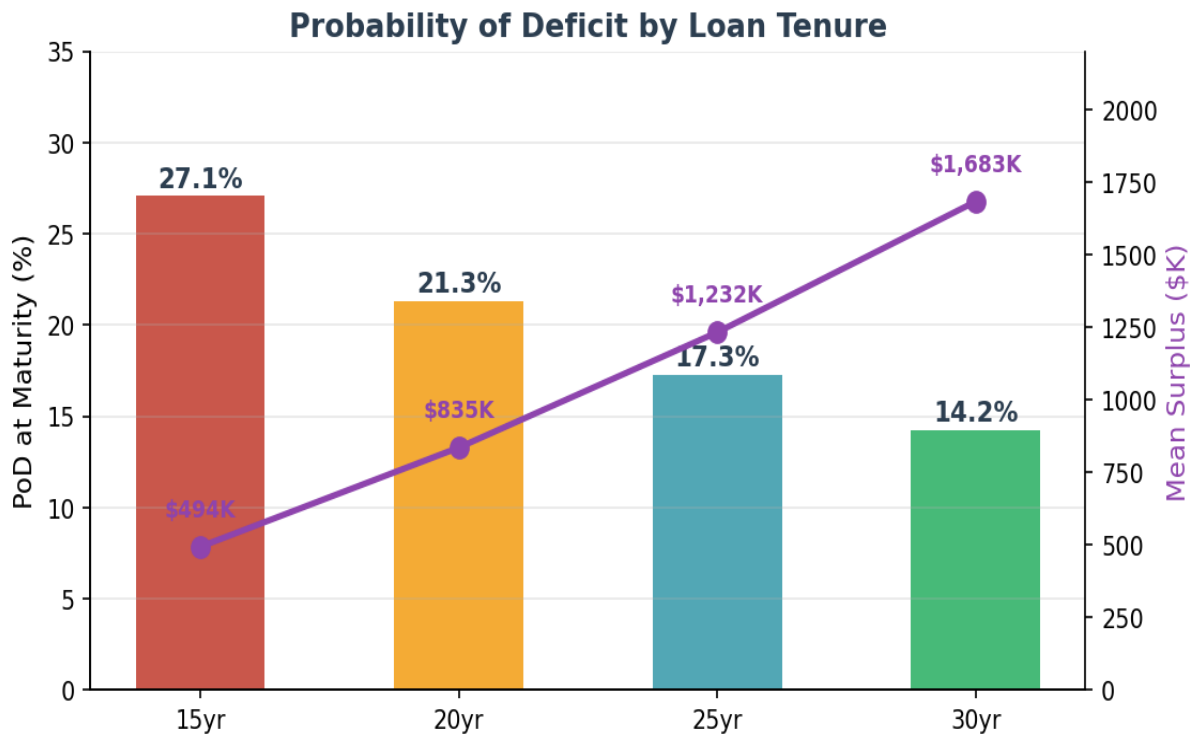
Scenario 2 — Top Cover Limit: Only the worst ~14% of paths are in deficit at Year 30. The P20 and P30 thresholds are already in surplus territory (+\$261K and +\$642K respectively), meaning a wider top cover beyond P10 would not engage LMI — the deficit is concentrated in the worst 14.25% of outcomes.

Model Parameters

Parameter	Value	Notes
Property Value	\$2,000,000	Reference property
LVR / Effective LVR	80% / 67.5%	Initial loan \$1,350,000
Annuity	\$25,000 p.a.	10-year annuity term
Equity Return	10% (vol 10%)	GBM, hedged $\pm 20\%$ collar
Cost of Capital	3.20%	Wholesale + retail + hedging + FP margin
Profit Share	25% every 5yr	Extracted from surplus, not at maturity
Simulation Paths	50,000 per tenure	Seed-controlled for reproducibility

1. Individual Tenure Analysis

Before comparing portfolio mixes, it is important to understand how each loan tenure performs individually. The key insight: **longer tenures consistently produce lower PoD and higher surplus**. This is because the investment has more time to compound past the cost burden.



Detailed Tenure Comparison

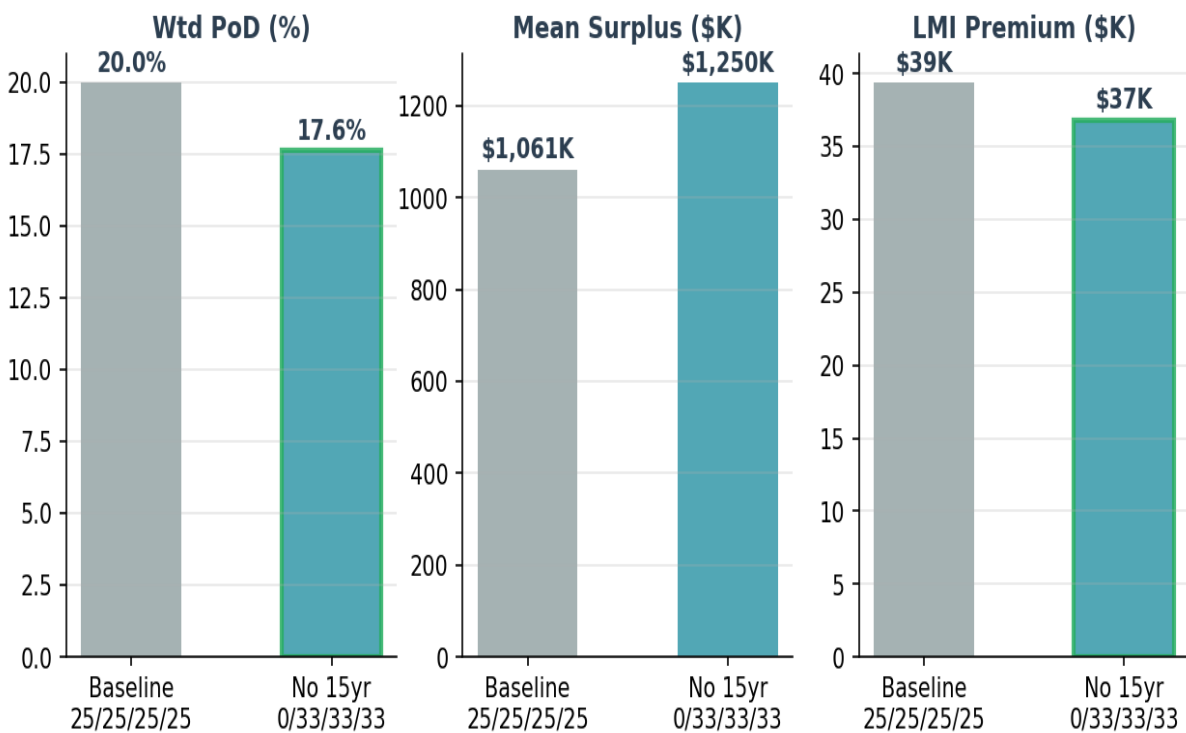
Metric	15yr	20yr	25yr	30yr
PoD at Maturity	27.09%	21.34%	17.27%	14.25%
Mean Surplus	\$493,646	\$835,141	\$1,232,227	\$1,682,642
Median Surplus	\$429,899	\$704,805	\$1,025,556	\$1,376,590
P10 (Worst 10%)	\$-485,989	\$-416,871	\$-337,581	\$-237,888
P20	\$-161,261	\$-38,632	\$100,440	\$260,824
P30	\$58,998	\$226,770	\$414,798	\$641,935

Key observation: PoD declines almost linearly with tenure — each additional 5 years of loan term reduces PoD by approximately 4-5 percentage points. The 15yr tenure has nearly **double** the PoD of the 30yr tenure (27.1% vs 14.3%). This is because shorter tenures have less compounding time for the investment to overcome the cost drag (3.20% p.a.) and generate surplus.

2. Scenario 1 — Removing 15-Year Terms

The current portfolio worksheet assumes an equal 25% allocation to each of 15, 20, 25, and 30-year loan terms. This scenario tests the impact of eliminating 15-year terms entirely and redistributing the allocation equally across the remaining three tenures.

Tenure	Baseline Weight	Test Weight	PoD at Maturity
15 years	25%	0%	27.09%
20 years	25%	33.3%	21.34%
25 years	25%	33.3%	17.27%
30 years	25%	33.3%	14.25%



Portfolio Impact

Metric	Baseline (25/25/25/25)	No 15yr (0/33/33/33)	Change
Weighted PoD	19.99%	17.62%	-2.37%
Portfolio PoC (Waterfall)	0.00%	0.00%	No change
Mean Surplus	\$1,060,914	\$1,250,003	+\$189,089
Loaded Premium	\$39,385	\$36,843	-\$2,542

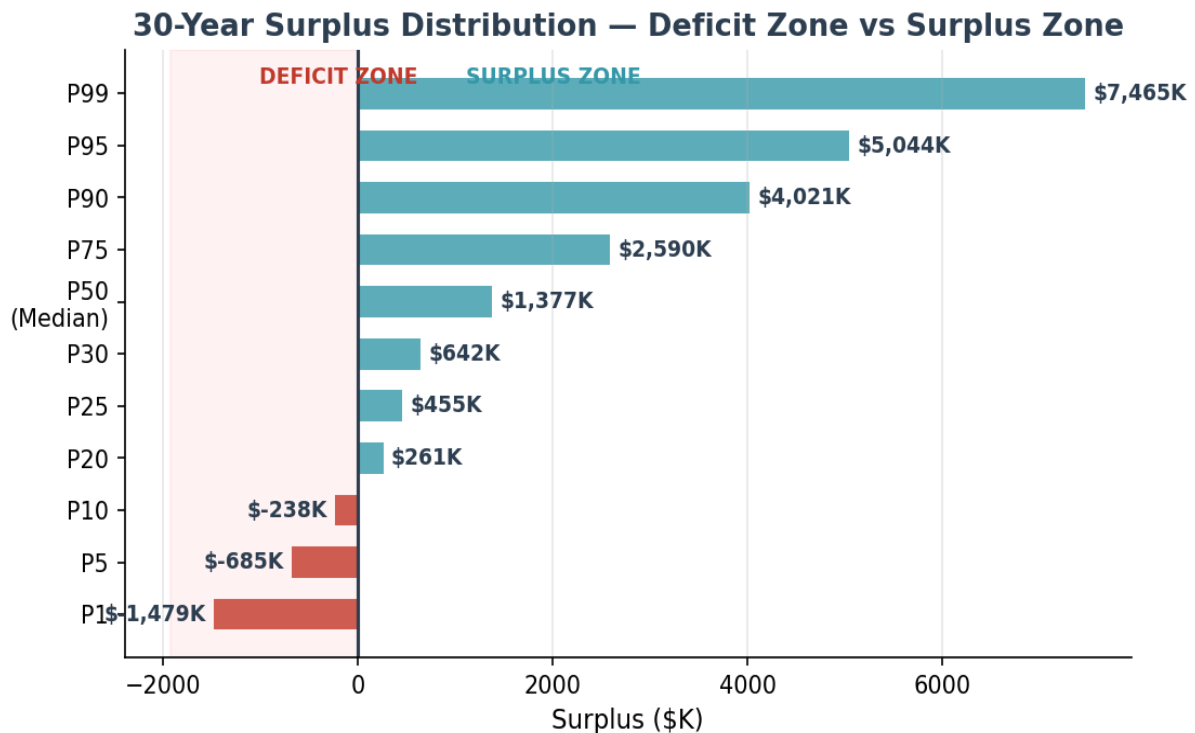
Conclusion — Scenario 1: Removing 15-year terms improves every metric. Weighted PoD drops by 2.4 percentage points, mean surplus increases by \$189K, and the insurance premium decreases by \$2,543 per mortgage. The intuition is confirmed: longer loan terms allow more compounding time, which more than compensates for the additional cost accrual. Portfolio PoC remains at 0.00% in both scenarios — the Payments Waterfall effectively eliminates residual claims at the portfolio level.

Recommendation: Consider restricting the minimum loan term to 20 years. This improves the risk profile with no adverse effect on the homeowner proposition — borrowers choosing shorter terms are likely less suited to the EPM product in any case, as they have shorter liquidity needs that conventional equity release products may serve better.

3. Scenario 2 — Top Cover Limit (LMI Threshold)

The current model sets the LMI top cover limit at the **worst 10% quantile (P10)** — the insurer covers losses up to the P10 threshold, and any excess goes to the reinsurance layer. This scenario tests what happens if we widen the top cover to P20 or P30.

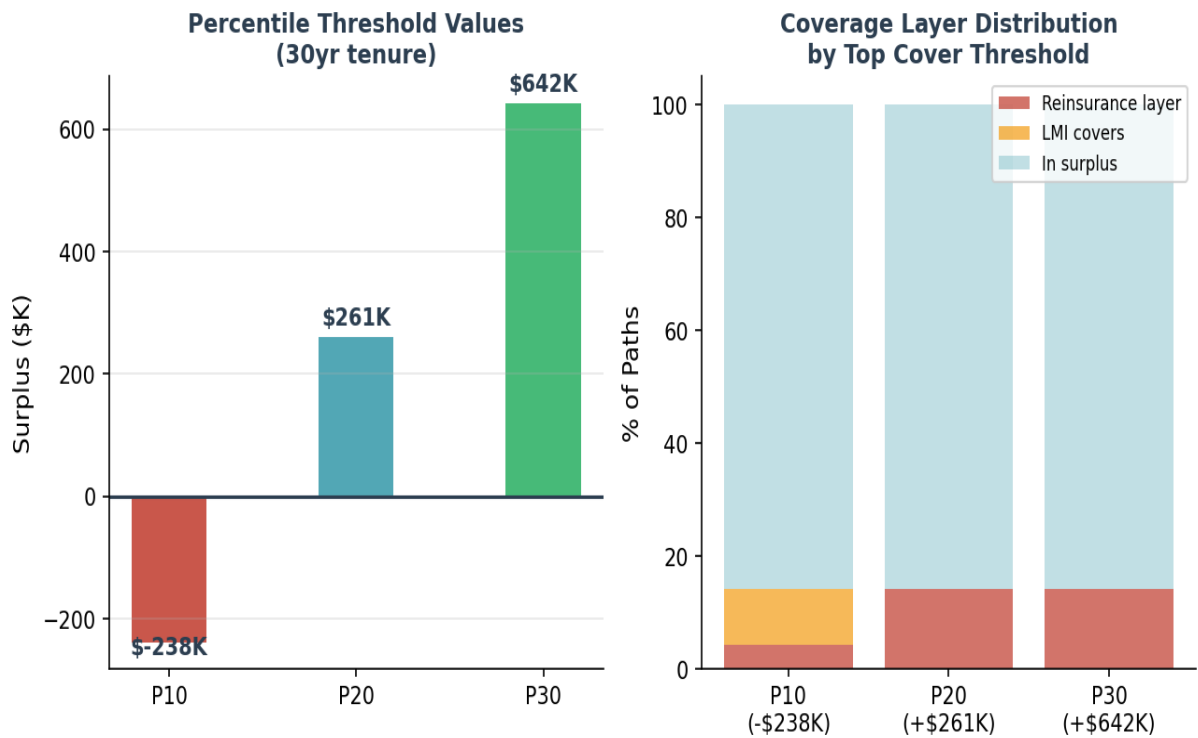
Surplus Distribution at Year 30



The chart above reveals the critical insight: **the P20 and P30 thresholds are already in surplus territory**. At Year 30, only 14.25% of paths are in deficit. The percentile thresholds are:

Percentile	Surplus Value	In Deficit?	Implication
P10 (worst 10%)	\$-237,888	Yes	LMI covers losses up to this amount
P20 (worst 20%)	\$260,824	No — surplus	No LMI engagement at this threshold
P30 (worst 30%)	\$641,935	No — surplus	No LMI engagement at this threshold

Insurance Layer Analysis (30yr tenure)



Top Cover	Threshold Value	LMI Premium (1.5x)	Reinsurance Premium	Paths Exceeding Cover
P10 (baseline)	\$237,888	\$11,463	\$22,740	10.00% (5,000 paths)
P20	\$0 (surplus)	\$0	\$34,203	14.25% (all deficit paths)
P30	\$0 (surplus)	\$0	\$34,203	14.25% (all deficit paths)

Conclusion — Scenario 2: Widening the top cover beyond P10 has **no practical effect** because the P20 and P30 thresholds are already in surplus. The deficit is concentrated in the worst 14.25% of paths, and the P10 threshold (-\$237,888) already captures the boundary between manageable and tail losses.

With P10 top cover: LMI absorbs losses for the 4.25% of paths between P10 and zero (premium: \$11,463), while the 10% of paths below P10 are covered by the reinsurance layer (\$22,740). Moving to P20/P30 would shift the entire cost burden to reinsurance (\$34,203) with no LMI layer at all — effectively eliminating the primary insurance tier.

Recommendation: The P10 top cover is well-positioned. It creates a natural split between LMI (covering moderate deficits) and reinsurance (covering tail risk). Widening to P20 or P30 would not improve the risk profile — it would simply remove the LMI layer entirely and place all risk into the reinsurance tier, which is typically more expensive per dollar of coverage.

4. Portfolio PoC & The Payments Waterfall

In both scenarios, the Portfolio PoC (Probability of Claim) simulated via the Payments Waterfall was **0.00%**. This requires explanation.

The Payments Waterfall operates by cross-subsidising deficit mortgages from surplus mortgages within the same portfolio. In a diversified 100-loan portfolio with vintage, tenure, and market entry point diversification, the surplus loans overwhelmingly compensate for the deficit loans:

Portfolio Metric	Value	Significance
Mean surplus per loan	\$1,060,914	Average across all tenures (baseline mix)
Deficit probability	~20%	Weighted average of individual tenures
Surplus-to-deficit ratio	~4:1	80% of loans in surplus vs 20% in deficit
Mean surplus of surplus loans	~\$1.5M	Ample buffer to cover deficit loans
Mean deficit of deficit loans	~\$470K	Much smaller than surplus available
Portfolio PoC	0.00%	No net portfolio deficit in 50K simulations

The 4:1 surplus-to-deficit ratio means that even in the worst portfolio simulations, the surplus loans generate enough cross-subsidisation to cover all deficit loans. This is why the Payments Waterfall is the **single most powerful risk mitigation mechanism** in the EPM framework — it reduces the effective claim rate from ~20% (individual PoD) to 0% (portfolio PoC).

5. Key Takeaways

- **Longer loan terms are unambiguously better:** Each additional 5 years of tenure reduces PoD by ~4-5 percentage points and increases mean surplus by ~\$300K. The 15yr tenure is the weakest performer at 27.09% PoD.
- **Removing 15yr terms improves every metric:** PoD drops 2.4pp, surplus rises \$189K, insurance premium falls \$2,543. There is no trade-off — the improvement is costless.
- **The P10 top cover is correctly positioned:** P20 and P30 are already in surplus, so widening the top cover has no practical effect. The deficit is concentrated in the worst 14.25% of paths — P10 captures the meaningful boundary.
- **Portfolio diversification eliminates claims:** The Payments Waterfall produces 0% PoC across all tested scenarios. The surplus-to-deficit ratio (~4:1) provides a substantial buffer against even correlated downturns.
- **Insurance primarily covers tail risk:** At P10, the LMI layer covers only \$11,463 per mortgage (0.72% of max loan). The reinsurance layer handles the extreme tail at \$22,740. Combined insurance cost is \$34,203 — approximately 2.1% of the max loan.

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